Deep Reinforcement Learning

Advanced Algorithms (Part I)

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

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Content

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- The evolution of modern RL algorithms
- Part (I): Improving policy gradient methods
  - Natural policy gradient
  - Trust-region policy optimization (TRPO)
  - Proximal policy optimization (PPO)

Where are we?

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	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
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Lecture contents

The evolution of modern RL algorithms

The evolution of modern RL algorithms

(I) Policy gradient methods

- Policy is explicitly parameterized and optimized directly:

$$L(\phi) = \mathbb{E}_{\tau \sim p_\phi(\tau)} [r(\tau)].$$

- Gradient descent: $\phi \leftarrow \phi + \alpha \nabla_\phi L(\phi)$.
- Methods are typically **on-policy**.
- **Algorithms**: REINFORCE $\rightarrow$ Actor-Critic $\rightarrow$ Natural Policy Gradient $\rightarrow$ Trust-region policy optimization (TRPO) $\rightarrow$ Proximal policy optimization (PPO).

Shortcomings

- High variance gradients.
- Unstable policy updates.
- Catastrophic performance collapse.

Improvement strategy

- How to safely update policies.

(II) Value-based methods

- Approximation of the Q-function:

$$Q^*(s, a) = r + \max_{a' \in \mathcal{A}} Q^*(s', a').$$

- Extract implicit policy: $\pi(s) = \arg\max_{a \in \mathcal{A}} Q^*(s, a)$.
- Typically **off-policy**.
- **Algorithms**: Q-learning $\rightarrow$ DQN $\rightarrow$ Deep deterministic policy gradient (DDPG) $\rightarrow$ TD3 $\rightarrow$ Soft actor-critic (SAC).

Shortcomings

- Instability from bootstrapping.
- Overestimation bias.
- Maximization over actions / continuous action spaces.

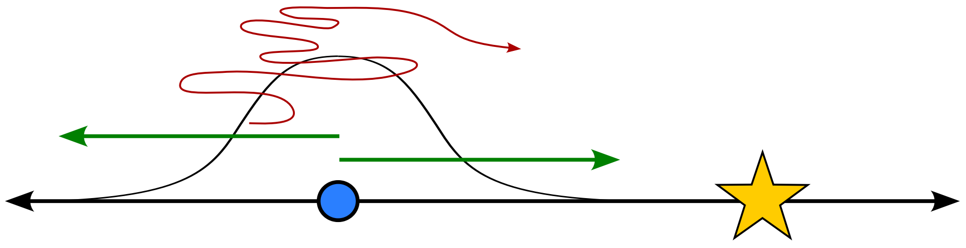
Improvement strategy

- Stabilizing Q-learning with function approximation.

- Solving the continuous argmax problem.

Part (I): Improving policy gradient methods

Ill-conditioned gradients

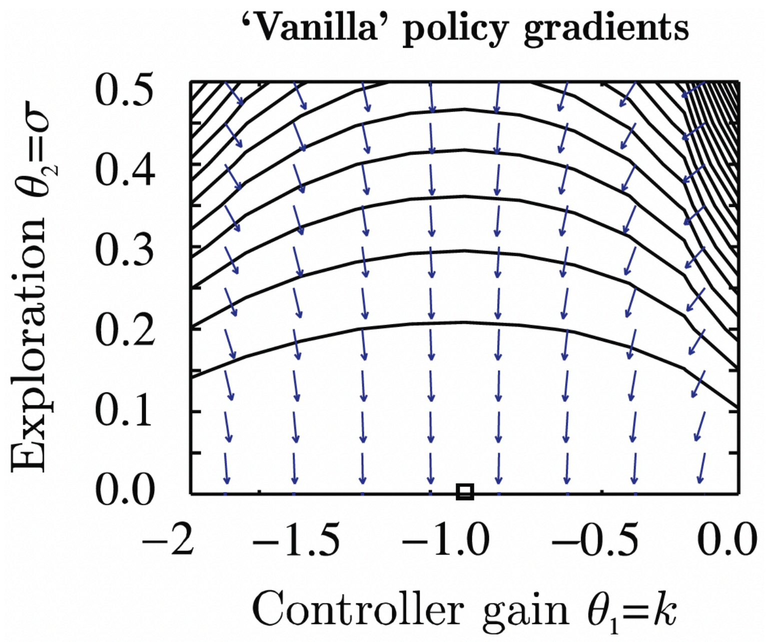


A simple exemplary MDP. Inspired by Sergey Levine's CS285 lecture.

$$r_t = -s_t^2 - a_t^2$$

$$\log \pi_\phi(a_t|s_t) = -\frac{1}{2\sigma^2}(ks_t - a_t)^2 + C, \qquad \phi = (k, \sigma).$$

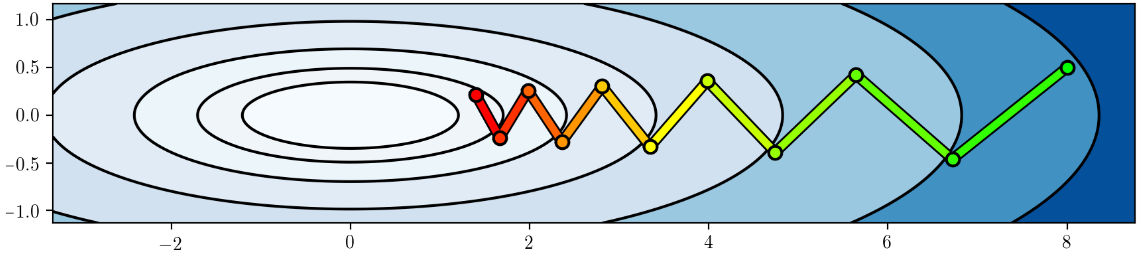
$$\nabla_\phi \log \pi_\phi(a_t|s_t) = \begin{pmatrix} -\frac{(ks_t - a_t)s_t}{\sigma^2} \\ \frac{(ks_t - a_t)^2}{\sigma^3} \end{pmatrix}$$



Source: (Peters and Schaal 2008)

- **Optimum**: $\phi^* = (-1, 0)$.
- **Issue**: The gradients do not point towards the optimum for smaller σ!
  - The σ^{-3} term blows up.

Closely related to ill-conditioned optimization problems:



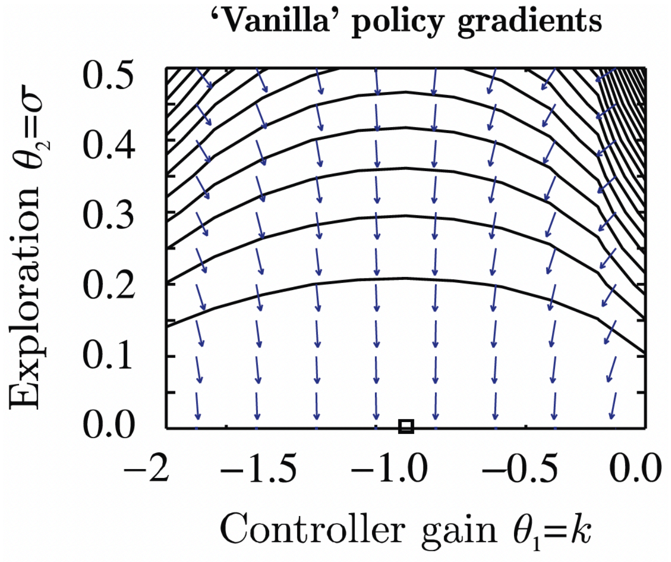
Constraining the ascent direction

- Recall the policy gradient update: $\phi \leftarrow \phi + \alpha \nabla_\phi L(\phi)$.
- A natural idea: **Constrain the length of our update step**!
  - Linearization of our optimization problem around ϕ using Taylor series expansion:

$$\begin{aligned} L(\phi') &= L(\phi) + \sum_{i=1}^{n} \frac{\partial L}{\partial \phi_i}(\phi'_i - \phi_i) + \mathcal{O}((\phi' - \phi)^2) \\ &= L(\phi) + (\phi' - \phi)^\top \nabla_\phi L(\phi). \end{aligned}$$

  - Constrained optimization of ϕ' along the steepest ascent direction:

$$\phi' \leftarrow \arg\max_{\phi'} (\phi' - \phi)^\top \nabla_\phi L(\phi) \qquad \text{subject to} \qquad \|\phi' - \phi\|^2 \leq \epsilon.$$



Source: (Peters and Schaal 2008)

- **But**: We just saw that some parameters change probabilities *a lot more than others*!
- **Alternative**: Rescale the gradient so that this does not happen!
- A better way to constrain the update distance: The **change in distribution** via the **Kullback-Leibler (KL) divergence**!

$$D_{\mathrm{KL}}\left(\pi_\phi \middle\| \pi'_\phi\right).$$

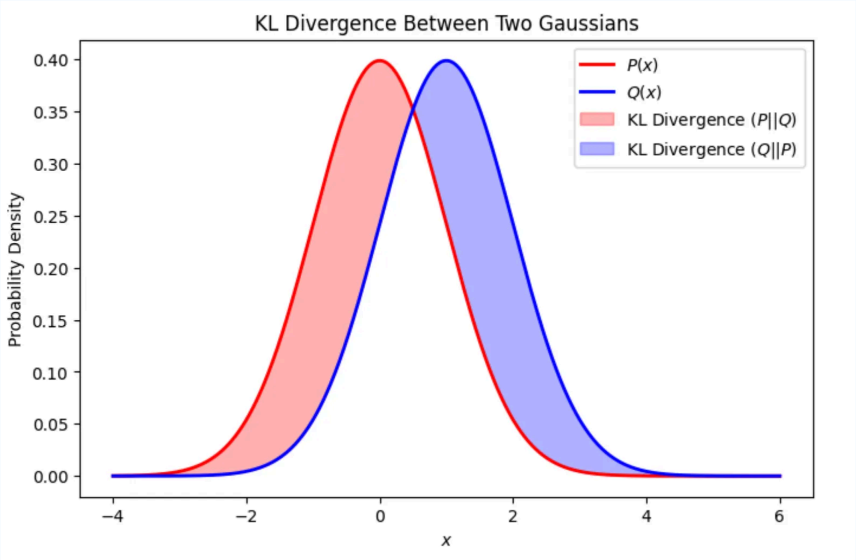
Aside: KL-divergence and Fisher information matrix

Kullback-Leibler divergence

- The **Kullback-Leibler divergence** (or **KL divergence**) measures the “distance” between two distributions (e.g., $p(x)$ and $q(x)$)

$$D_{\mathsf{KL}}\left(p \| q\right) = \begin{cases} \sum_{x \in \mathcal{X}} p(x) \log\left(\frac{p(x)}{q(x)}\right) & \text{if } \mathcal{X} \text{ is finite} \\ \int \mathcal{X} p(x) \log\left(\frac{p(x)}{q(x)}\right) \mathrm{d}x & \text{if } \mathcal{X} \text{ is continuous} \end{cases}.$$

  - It is zero if and only if $p(x) = q(x)$.
- **Intuition**: Imagine you have two different maps of the exact same city.
  - Map A is completely accurate. Every street, etc., is exactly where it should be.
  - Map B was drawn from memory by a friend. It’s mostly right with some minor flaws.
  - If you use Map B to navigate the city, you are going to make some mistakes.
  - The KL divergence answers the question: “How much extra ‘gas’ (or information) will I waste if I use Map B instead of Map A?”



[*Source*].

- KL Divergence measures the *surprise* or *unfair penalty* you get when you use q to predict something that actually follows p. If q is a perfect match for p (i.e., $q = p$), your surprise is zero. The worse your guess (q) is, the higher the KL divergence.
- **Question**: What happens if our “model” q is absolutely certain that something is **not** going to happen ($q(x) = 0$ for some x)?

💡 The KL divergence is not a real distance. For instance, it it not symmetric (i.e., $D_{\mathsf{KL}}\left(p \| q\right) \neq D_{\mathsf{KL}}\left(q \| p\right)$ in general), which means that it also does not satisfy the triangle inequality. Still, it gives us a measure of how far two distributions are apart.

Fisher information matrix

- The **Fisher information matrix** (**FIM**) measures how much information an observable random variable x carries about an unknown parameter ϕ (e.g., the weights of a neural network).
- Mathematically, if we have a log-likelihood function $\log p(x|\phi)$, the FIM (denoted as F) is the variance of its gradient (the "*score*"):

$$F = \mathbb{E}\left[(\nabla_\phi \log p(x|\phi))(\nabla_\phi \log p(x|\phi))^\top\right]. \tag{1}$$

In simpler terms: How do variations of ϕ influence the resulting probability distribution?

- High Fisher Information: Small change in $\phi \Rightarrow$ large shift in the distribution.
- Low Fisher Information: Change in $\phi \Rightarrow$ only small shift in the distribution.

Relation to the KL divergence

- Consider a very small change in ϕ by $\Delta\phi$, and study the KL divergence $D_{\text{KL}}(p(x|\phi) \| p(x|\phi + \Delta\phi))$.
- If we perform a Taylor series expansion of D_{KL} around $\Delta\phi = 0$, then the first two terms are zero ($D_{\text{KL}}(p(x|\phi) \| p(x|\phi))$ $= 0$ and since $\Delta\phi = 0$ is the minimum, the first derivative is zero as well).
- The second order term is thus the leading term! $\Rightarrow$ for small $\Delta\phi$, we have

$$D_{\text{KL}}(p(x|\phi) \| p(x|\phi + \Delta\phi)) \approx \frac{1}{2}\Delta\phi^\top F \Delta\phi. \tag{2}$$

KL-divergence and Fisher information in RL

- When using the KL divergence or the Fisher information matrix in RL, we often do so in terms of different policies.
- This means that we consider the difference in the **action distribution** $D_{\mathsf{KL}}\left(\pi_\phi \| \pi_{\phi'}\right)$.
- However, this expression is incomplete or even misleading.
- Instead, we consider either

$$D_{\mathsf{KL}}\left(\pi_\phi(\cdot|s) \| \pi_{\phi'}(\cdot|s)\right) = \int_{\mathcal{A}} \pi_\phi(a|s) \log\left(\frac{\pi_\phi(a|s)}{\pi_{\phi'}(a|s)}\right) \mathrm{d}a$$

  for a fixed s, or the expectation over the state space, i.e.,

$$\mathbb{E}_{s\sim p_\phi}\left[D_{\mathsf{KL}}\left(\pi_\phi(\cdot|s) \| \pi_{\phi'}(\cdot|s)\right)\right].$$

- The relation between KL divergence and FIM for small changes can be derived analogously to the previous slide and reads

$$\mathbb{E}_{s\sim p_\phi}\left[D_{\mathsf{KL}}\left(\pi_\phi \| \pi_{\phi'}\right)\right] \approx \frac{1}{2}\Delta\phi^\top \Big(\underbrace{\mathbb{E}_{s\sim p_\phi}[F(s)]}_{=F}\Big)\Delta\phi = \frac{1}{2}\Delta\phi^\top F \Delta\phi.$$

Natural policy gradient

Back to constrained ascent directions

- Recall: constraining w.r.t. the parameter ϕ does not solve the issue of strongly different impacts of individual parameters:

$$\phi' \leftarrow \arg\max_{\phi'} (\phi' - \phi)^\top \nabla_\phi L(\phi) \qquad \text{subject to} \qquad \|\phi' - \phi\|^2 \leq \epsilon.$$

- **Question**: What is a good alternative to avoid too large (and thus, harmful) steps?
$\Rightarrow$ the KL divergence!

$$\phi' \leftarrow \arg\max_{\phi'} (\phi' - \phi)^\top \nabla_\phi L(\phi) \qquad \text{subject to} \qquad \mathbb{E}_{s \sim p_\phi} \left[D_{\mathsf{KL}} \left(\pi_{\phi'}(\cdot|s) \| \pi_\phi(\cdot|s) \right) \right] \leq \epsilon.$$

  - Constraining the KL divergence means that we automatically treat different parameters differently!
  - If a change in one of the entries of ϕ results in large policy changes, then the constraint does not allow large changes in that entry.

- Since we want to have small updates (constrained by ϵ): approximation via the Fisher information matrix,

$$\phi' \leftarrow \arg\max_{\phi'} (\phi' - \phi)^\top \nabla_\phi L(\phi) \qquad \text{subject to} \qquad (\phi' - \phi)^\top F (\phi' - \phi),$$

$$\text{with} \qquad F = \mathbb{E}_{s \sim p_\phi, a \sim \pi_\phi(\cdot|s)} \left[(\nabla_\phi \log \pi_\phi(a|s))(\nabla_\phi \log \pi_\phi(a|s))^\top \right].$$

Natural policy gradient

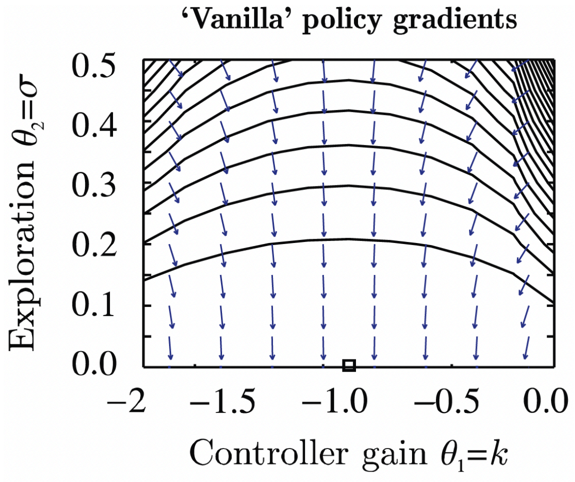
- We have now defined a policy gradient whose update length is constrained in terms of the KL divergence:

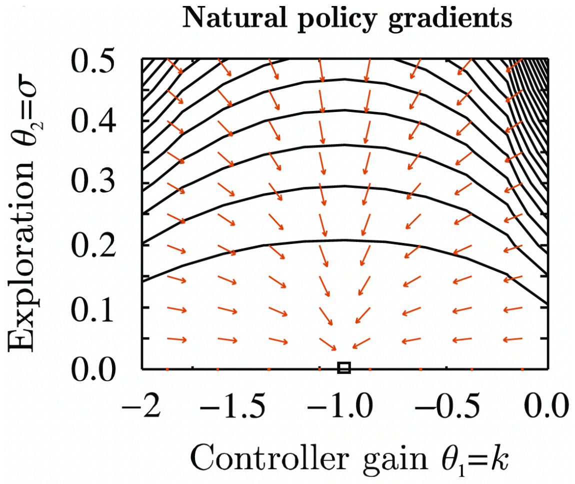
$$\phi' \leftarrow \arg\max_{\phi'} (\phi' - \phi)^\top \nabla_\phi L(\phi) \quad \text{subject to} \quad (\phi' - \phi)^\top F(\phi' - \phi). \tag{3}$$

- What remains is the question of solving (3).
- Without going into further details (e.g., (S. M. Kakade 2001; Peters and Schaal 2008))
  $\Rightarrow$ Solution: **scale** the policy gradient **by the inverse Fisher information matrix**,

$$\phi \leftarrow \phi + \alpha F^{-1} \nabla_\phi L(\phi). \tag{4}$$

- **Intuition** behind the inverse FIM F^{-1}:
  - Parameters with a high impact have high Fisher information.
  - Scaling by the inverse “normalizes” the individual impacts.
- This techinque is known as the **natural policy gradient** (also **covariant policy gradient**).
  - Preconditioning steps of this type are very common in optimization in general.





Source: (Peters and Schaal 2008)

- Drawback: $F \in \mathbb{R}^{d \times d}$ can be very expensive to calculate (and invert) for very large parameter vectors.

Limitations of the natural policy gradient

$$\phi \leftarrow \phi + \alpha F^{-1} \nabla_\phi L(\phi) \tag{4}$$

1. The step size α is hard to choose!

- Scaling doesn't prevent the step from changing the policy too much.
- Large steps can collapse performance.

2. No explicit control of policy change.

- NPG does not explicitly constrain the distance between π_{old} and π_{new}.
- Data for gradient estimation is sampled from π_{old}.
- If new policy differs too much, gradient estimate becomes unreliable.

3. Inversion of the Fisher matrix.

- NPG requires inversion of the Fisher Information Matrix.
- If our network has d parameters, the FIM is a $d \times d$ matrix.
- For a modest deep learning network with 100,000 parameters, F contains 10 billion elements.

4. Performance guarantees require small policy changes.

- NPG does not provide a mathematical guarantee that π_{new} will actually be better than (or at least as good as) π_{old}.
- Theory shows that small policy updates should improve performance (S. Kakade and Langford 2002) …
- … but the basic NPG update does not enforce the required small change condition.

- Storing this in memory and computing its exact inverse ($\mathcal{O}(d^3)$ complexity) $\Rightarrow$ computationally impossible.

Trust-region policy optimization

Performance measures of policies

- Let's introduce a new (and yet *already known* quantity): the **performance measure** of a policy,

$$\eta(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T-1}\gamma^t r_t\right] = \mathbb{E}_{s_0\sim p_0}\left[V^\pi(s_0)\right].$$

- Remember our reinforcement learning objective

$$L(\phi) = \mathbb{E}_{\tau\sim p_\phi(\tau)}\left[\sum_{t=0}^{T-1} r_t\right] = \mathbb{E}_{s_0\sim p_0}\left[V^{\pi_\phi}(s_0)\right]?$$

- If we consider a policy parameterized by ϕ, i.e., π_ϕ, then the two equations are the same,

$$\eta(\pi_\phi) = L(\phi).$$

- The difference is more in terms of the *motivation*, not the definition:
  - We use $L(\phi)$ in policy gradients, since we want to find ascent directions for the network paraemters.
  - We use $\eta(\pi)$ as a global quantity to establish mathematical lower-bounds so that an algorithm can compute a safe step size (the trust region).

💡 For more details on this matter, see (S. Kakade and Langford 2002; Schulman et al. 2015).

Comparing the performance measure of two policies (1)

- Can we relate two policies π_{old} and π_{new} using advantage functions?
- Let's use $\gamma = 1$ for simplicity (but one can work this out with γ as well).
- Consider the advantage function under the old policy π_{old},

$$A_{\pi_{\text{old}}}(s_t, a_t) = r_t + V^{\pi_{\text{old}}}(s_{t+1}) - V^{\pi_{\text{old}}}(s_t).$$

- Now sample a trajectory under the new policy π_{new} (i.e., $(s_0, a_0) \to \ldots \to (s_{T-1}, a_{T-1})$) and calculate the advantages:

$$\begin{aligned}
A_{\pi_{\text{old}}}(s_0, a_0) &= r_0 + V^{\pi_{\text{old}}}(s_1) - V^{\pi_{\text{old}}}(s_0) \\
A_{\pi_{\text{old}}}(s_1, a_1) &= r_1 + V^{\pi_{\text{old}}}(s_2) - V^{\pi_{\text{old}}}(s_1) \\
&\vdots \\
A_{\pi_{\text{old}}}(s_{T-2}, a_{T-2}) &= r_{T-2} + V^{\pi_{\text{old}}}(s_{T-1}) - V^{\pi_{\text{old}}}(s_{T-2}) \\
A_{\pi_{\text{old}}}(s_{T-1}, a_{T-1}) &= r_{T-1} + \underbrace{V^{\pi_{\text{old}}}(s_T)}_{=0 \text{ (terminal)}} - V^{\pi_{\text{old}}}(s_{T-1})
\end{aligned}$$

Comparing the performance measure of two policies (2)

- Take the sum:

$$\sum_{t=0}^{T-1} A_{\pi_{\text{old}}}(s_t, a_t) = \underbrace{r_0 + V^{\pi_{\text{old}}}(s_1) - V^{\pi_{\text{old}}}(s_0)}_{=A_{\pi_{\text{old}}}(s_0, a_0)} + \underbrace{r_1 + V^{\pi_{\text{old}}}(s_2) - V^{\pi_{\text{old}}}(s_1)}_{=A_{\pi_{\text{old}}}(s_1, a_1)} + \ldots + \underbrace{r_{T-2} + V^{\pi_{\text{old}}}(s_{T-1}) - V^{\pi_{\text{old}}}(s_{T-2})}_{=A_{\pi_{\text{old}}}(s_{T-2}, a_{T-2})} + \underbrace{r_{T-1} - V^{\pi_{\text{old}}}(s_{T-1})}_{=A_{\pi_{\text{old}}}(s_{T-1}, a_{T-1})}$$

$$= r_0 + r_1 + \ldots + r_{T-2} + r_{T-1} - V^{\pi_{\text{old}}}(s_0) = \left(\sum_{t=0}^{T-1} r_t\right) - V^{\pi_{\text{old}}}(s_0).$$

- Shift around and take the expectation with respect to the new policy (in the case of $V^{\pi_{\text{old}}}$, this reduces to $s_0 \sim p_0$):

$$\mathbb{E}_{s \sim p_{\pi_{\text{new}}}, a \sim \pi_{\text{new}}}\left[\left(\sum_{t=0}^{T-1} r_t\right)\right] = \mathbb{E}_{s_0 \sim p_0}\left[V^{\pi_{\text{old}}}(s_0)\right] + \mathbb{E}_{s \sim p_{\pi_{\text{new}}}, a \sim \pi_{\text{new}}}\left[\sum_{t=0}^{T-1} A_{\pi_{\text{old}}}(s_t, a_t)\right]$$

$$\Leftrightarrow \qquad \eta(\pi_{\text{new}}) = \eta(\pi_{\text{old}}) + \mathbb{E}_{s \sim p_{\pi_{\text{new}}}, a \sim \pi_{\text{new}}}\left[\sum_{t=0}^{T-1} \gamma^t A_{\pi_{\text{old}}}(s_t, a_t)\right]$$

- Right now, the expectation is written over an entire lifetime trajectory ($\tau \sim \pi_{\text{new}}$).
- To be useful for an optimization algorithm, we need to break it down step-by-step and look at individual states and actions.

Comparing the performance measure of two policies (3)

- Let's consider the continual case ($T=\infty$) such that we can disect the expectation into individual steps:

$$
\begin{aligned}
\mathbb{E}_{s\sim p_{\pi_\text{new}},a\sim\pi_\text{new}}\left[\sum_{t=0}^{\infty}\gamma^t A_{\pi_\text{old}}(s_t,a_t)\right] &= \sum_{t=0}^{\infty}\gamma^t \mathbb{E}_{s\sim p_{\pi_\text{new}},a\sim\pi_\text{new}}\left[A_{\pi_\text{old}}(s_t,a_t)\right] \\
&= \sum_{t=0}^{\infty}\int_{\mathcal{S}} p_{\pi_\text{new}}(s)\int_{\mathcal{A}}\pi_\text{new}(a|s)A_{\pi_\text{old}}(s,a)\,\mathrm{d}a\,\mathrm{d}s \\
&= \int_{\mathcal{S}}\underbrace{\left(\sum_{t=0}^{\infty}\gamma^t p_{\pi_\text{new}}(s)\right)}_{=\rho_{\pi_\text{new}}(s)}\left(\int_{\mathcal{A}}\pi_\text{new}(a|s)A_{\pi_\text{old}}(s,a)\,\mathrm{d}a\right)\mathrm{d}s \\
&= \mathbb{E}_{s\sim\rho_{\pi_\text{new}},a\sim\pi_\text{new}}\left[A_{\pi_\text{old}}(s,a)\right].
\end{aligned}
$$

- Here, we have introduced the **discounted state distribution** $\rho_{\pi_\text{new}}(s)$.

Relation between the performance measures

$$
\eta(\pi_\text{new}) = \eta(\pi_\text{old}) + \mathbb{E}_{s\sim\rho_{\pi_\text{new}},a\sim\pi_\text{new}}\left[A_{\pi_\text{old}}(s,a)\right] \tag{5}
$$

💡 When sampling, the data will automatically be distributed according to $\rho_{\pi_\text{new}}(s)$. See also the discussion in the actor-critic lecture.

Comparing the performance measure of two policies (4)

Relation between the performance measures

$$\eta(\pi_{\text{new}}) = \eta(\pi_{\text{old}}) + \mathbb{E}_{s\sim\rho_{\pi_{\text{new}}}, a\sim\pi_{\text{new}}}\left[A_{\pi_{\text{old}}}(s,a)\right] \tag{5}$$

What's so surprising about this equation?

- The Advantage function is built entirely out of the old policy's values,
- yet when we weight it by the new policy's state distribution,
- it perfectly calculates the new policy's total return.

Upside and downside of this identity

$+$ Proof that if you can find a new policy with a positive expected advantage under the old policy, your policy will get better.

— To compute the exact value, we must sample states from the new policy ($s \sim p_{\pi_{\text{new}}}$), which we do not yet know!

Solution approach in TRPO: "If π_{new} is very close to π_{old} (enforced by a KL-constraint), we can swap the state distribution $p_{\pi_{\text{new}}}$ for $p_{\pi_{\text{new}}}$ and safely optimize an approximation (i.e., a *surrogate objective*)."

How to turn this into an algorithm

Relation between the performance measures

$$\eta(\pi_{\text{new}}) = \eta(\pi_{\text{old}}) + \mathbb{E}_{s\sim\rho_{\pi_{\text{new}}},a\sim\pi_{\text{new}}}\left[A_{\pi_{\text{old}}}(s,a)\right] \tag{5}$$

Starting with (5), we need to take the several steps to arrive at an algorithm that we can actually implement.

1. Construct a local approximation of (5) (the *surrogate loss*) which we can estimate using samples.
2. Ensure that this approximation is sufficiently accurate.
3. Turn he surrogate loss function into a practical optimization problem.
4. Make sure that it scales to large problems.

Trust-region policy optimization (TRPO)
The first realization of these steps resulted in the TRPO algorithm, which was derived in (Schulman et al. 2015).

TRPO: The surrogate loss function

Relation between the performance measures

$$\eta(\pi_{\text{new}}) = \eta(\pi_{\text{old}}) + \mathbb{E}_{s\sim\rho_{\pi_{\text{new}}},a\sim\pi_{\text{new}}}\left[A_{\pi_{\text{old}}}(s,a)\right] \tag{5}$$

- In principle, this equation shows us a way to guarantee policy improvement.
- If we can ensure positive expected advantages, then we improve!
- **But**: Sampling under the new policy ($s \sim \rho_{\pi_{\text{new}}}, a \sim \pi_{\text{new}}$) is infeasible, as we do not have it yet!

A surrogate loss function

- Simple trick: just sample the state s under the *old* policy!

$$L_{\pi_{\text{old}}}(\pi) = \eta(\pi_{\text{old}}) + \mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi}\left[A_{\pi_{\text{old}}}(s,a)\right]. \tag{6}$$

- Under the assumption that π_{old} and π_{new} do not differ too much, this is a reasonable assumption.
- In the limit case $\pi_{\text{old}} = \pi_{\text{new}}$, we have equality of (5) and (6), as well as their gradients (Schulman et al. 2015, Eq. (4)).

TRPO: Approximation error

In (Schulman et al. 2015), it was shown that error between the two expressions (5) and (6) can be bounded

$$\eta(\pi) \geq L_{\pi_{\text{old}}}(\pi) - C \cdot D_{\text{KL}}^{\max}(\pi_{\text{old}} \| \pi), \qquad \text{with} \quad D_{\text{KL}}^{\max}(\pi_{\text{old}} \| \pi) = \max_{s \in \mathcal{S}} D_{\text{KL}}(\pi_{\phi}(\cdot|s) \| \pi_{\phi'}(\cdot|s)),$$

and C is a constant derived from the environment's transition dynamics and the maximum bounds of the advantage function.

Central message: If we maximize the right-hand side of this inequality, we are guaranteed to improve (or at least maintain) our true performance $\eta(\pi)$!

Maximizing the surrogate loss $\Rightarrow$ introducing a parametric policy π_ϕ, our surrogate-based optimization problem is

$$\max_{\phi} L_{\pi_{\text{old}}}(\pi_\phi) - C \cdot D_{\text{KL}}^{\max}(\pi_{\text{old}} \| \pi_\phi),$$

or, if we consider two iterates ϕ and ϕ'

$$\max_{\phi'} L_{\pi_\phi}(\pi_{\phi'}) - C \cdot D_{\text{KL}}^{\max}(\pi_\phi \| \pi_{\phi'}).$$

Challenges:

1. The constant C is usually large and thus restrictive

   Solution: Transfer to a constraint with hyperparameter δ.

2. Computing the maximum KL divergence over all possible states ($D_{\text{KL}}^{\max}(\pi_{\phi_{\text{old}}} \| \pi_\phi)$) is impossible.

   Solution: Replace by the average KL divergence over the states that were actually observed:

$$\bar{D}_{\text{KL}}(\pi_\phi \| \pi_{\phi_{\text{old}}}) = \mathbb{E}_{s \sim \rho_{\pi_{\phi_{\text{old}}}}}\left[D_{\text{KL}}(\pi_{\phi_{\text{old}}}(\cdot|s) \| \pi_\phi(\cdot|s))\right].$$

TRPO: Sampling the correct expectation

Our final observation is that in our surrogate loss function $L_{\pi_{\text{old}}}$ (Eq. (6)), the "old" performance measure $\eta(\pi_{\text{old}})$ serves as a constant $\Rightarrow$ we can neglect it!

- Overall, we obtain the following surrogate optimization problem:

$$\max_{\phi} \underbrace{\mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_\phi}[A_{\pi_{\text{old}}}(s,a)]}_{=L_{\pi_{\text{old}}}(\pi_\phi)-\eta(\pi_{\text{old}})} \quad \text{subject to} \quad \underbrace{\mathbb{E}_{s\sim\rho_{\pi_{\text{old}}}}[D_{\text{KL}}(\pi_{\text{old}}(\cdot|s)\|\pi_\phi(\cdot|s))]}_{=\bar{D}_{\text{KL}}(\pi_{\text{old}}\|\pi_\phi)} \leq \delta.$$

- What's wrong with this equation?
  $\Rightarrow$ In the loss, we're compute expectations over the new policy π_ϕ, but the data was collected under π_{old}!

$$\mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_\phi}\left[A_{\pi_{\text{old}}}(s,a)\right] = \int_{\mathcal{S}} \rho_{\pi_{\text{old}}}(s) \int_{\mathcal{A}} \pi_\phi(a|s) A_{\pi_{\text{old}}}(s,a)\,\mathrm{d}a\,\mathrm{d}s.$$

- **Solution**: Importance sampling!

$$\int_{\mathcal{S}} \rho_{\pi_{\text{old}}}(s) \int_{\mathcal{A}} \pi_{\text{old}}(a|s) \frac{\pi_\phi(a|s)}{\pi_{\text{old}}(a|s)} A_{\pi_{\text{old}}}(s,a)\,\mathrm{d}a\,\mathrm{d}s = \mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_{\text{old}}}\left[\frac{\pi_\phi(a|s)}{\pi_{\text{old}}(a|s)} A_{\pi_{\text{old}}}(s,a)\right].$$

TRPO: The surrogate optimization problem

Trust-region policy optimization (TRPO) optimization problem

$$\max_{\phi} \underbrace{\mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_{\text{old}}}\left[\frac{\pi_\phi(a|s)}{\pi_{\text{old}}(a|s)}A_{\pi_{\text{old}}}(s,a)\right]}_{=L_{\text{TRPO}}(\phi)} \quad \text{subject to} \quad \underbrace{\mathbb{E}_{s\sim\rho_{\pi_{\text{old}}}}[D_{\text{KL}}(\pi_{\text{old}}(\cdot|s)\|\pi_\phi(\cdot|s))]}_{=\bar{D}_{\text{KL}}(\pi_{\text{old}}\|\pi_\phi)} \leq \delta. \quad (7)$$

Next question: How do we solve it efficiently? $\Rightarrow$ Let's start with the gradient!

$$\nabla_\phi L_{\text{TRPO}}(\phi) = \nabla_\phi \mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_{\text{old}}}\left[\frac{\pi_\phi(a|s)}{\pi_{\text{old}}(a|s)}A_{\pi_{\text{old}}}(s,a)\right] = \mathbb{E}_{s\sim\rho_{\pi_{\text{old}}},a\sim\pi_{\text{old}}}\left[\frac{\nabla_\phi \pi_\phi(a|s)}{\pi_{\text{old}}(a|s)}A_{\pi_{\text{old}}}(s,a)\right]$$

The Policy gradient for the TRPO surrogate loss

$$
\begin{aligned}
g = \nabla_\phi L_{\text{TRPO}}(\phi)\Big|_{\color{red}{\phi=\phi_{\text{old}}}} &= \mathbb{E}_{s\sim\rho_{\pi_{\phi_{\text{old}}}}, a\sim\pi_{\phi_{\text{old}}}}\left[\frac{\nabla_\phi \pi_\phi(a|s)\big|_{\phi=\phi_{\text{old}}}}{\pi_{\phi_{\text{old}}}(a|s)} A_{\pi_{\phi_{\text{old}}}}(s,a)\right] \\
&= \mathbb{E}_{s\sim\rho_{\pi_{\phi_{\text{old}}}}, a\sim\pi_{\phi_{\text{old}}}}\left[\nabla_\phi \log \pi_\phi(a|s)\big|_{\phi=\phi_{\text{old}}} A_{\pi_{\phi_{\text{old}}}}(s,a)\right] \quad \textit{(Log-identity trick)}
\end{aligned} \tag{8}
$$

is the standard policy gradient with value baseline!

TRPO: Solving the optimization problem (7)

To solve Problem (7) efficiently, we need two steps:

1. Linearize the objective function: First-order Taylor series expansion around ϕ_{old} using the gradient g (Eq. (8)).

$$L_{\text{TRPO}}(\phi) \approx L_{\text{TRPO}}(\phi_{\text{old}}) + g^\top(\phi - \phi_{\text{old}}) = g^\top(\phi - \phi_{\text{old}}).$$

   (The term $L_{\text{TRPO}}(\phi_{\text{old}})$ drops out because it is exactly zero at ϕ_{old}: No advantage and sampling ratio is one.)

2. Quadratic approximation of the KL constraint for small $\Delta\phi = (\phi - \phi_{\text{old}}) \Rightarrow$ Fisher information matrix (Eq. (2))!

Linear-quadratic approximation of the TRPO problem (7)

$$\max_{\Delta\phi} g^\top \Delta\phi \quad \text{subject to} \quad \frac{1}{2}\Delta\phi^\top F \Delta\phi \leq \delta, \tag{9}$$

where, according to (1),

$$F(\phi) = \mathbb{E}_{s\sim\rho_{\pi_{\phi_{\text{old}}}}, a\sim\pi_{\phi_{\text{old}}}}\left[\left[(\nabla_\phi \log \pi_\phi(a|s))(\nabla_\phi \log \pi_\phi(a|s))^\top\right]_{\phi=\phi_{\text{old}}}\right].$$

Solution to (9)
The solution (via Lagrange multipliers) yields

$$\Delta\phi = \sqrt{\frac{2\delta}{g^\top F^{-1} g}} F^{-1} g. \tag{10}$$

Note: This is NPG (4) with automatic step size selection $\alpha = \sqrt{\frac{2\delta}{g^\top F^{-1} g}}$!

💡 In classical optimization, the local approximation (9) is solved repeatedly. The solution can be shown to converge to the optimum of (7), which is known as **Sequential Quadratic Programming** (**SQP**). In TRPO, we solve the approximation (9) just once before collecting new data.

TRPO: Avoiding matrix inversion

$$\Delta\phi = \sqrt{\frac{2\delta}{g^\top F^{-1} g}} F^{-1} g \tag{10}$$

- Matrix inversion scales cubically. That is, for a d-dimensional parameter $\phi \in \mathbb{R}^d$, we have $\mathcal{O}(d^3)$.
- Solution: Use the conjugate gradient (CG) method for solving linear systems of the form $Fx = g$.
  - Usually, CG scales according to $\mathcal{O}(d^2)$ per iteration.
  - However, in TRPO, we can avoid calculating $F \in \mathbb{R}^{d\times d}$ entirely using automatic differentiation $\Rightarrow$ scales linearly with the number of weights (i.e., $\mathcal{O}(d)$).
  - This is as fast as it gets!

Approach

- Compute $x = F^{-1} g$ using the CG method ($\mathcal{O}(d)$ if we use autodiff).
- Calculate the denominator in (10) (Note that $F^\top = F$, since the FIM is symmetric):

$$\nu = g^\top F^{-1} g = (Fx)^\top F^{-1} Fx = x^\top F^\top F^{-1} Fx = x^\top Fx = x^\top g.$$

- Compute the step size $\beta = \sqrt{\frac{2\delta}{\nu}} F^{-1} g$.

- Compute the update $\Delta\phi = \beta x$.

TRPO: Step size selection

TRPO surrogate problem

$$\max_{\phi} L_{\text{TRPO}}(\phi) \quad \text{s.t.} \quad \bar{D}_{\text{KL}}\left(\pi_{\text{old}} \| \pi_{\phi}\right) \leq \delta. \qquad (7)$$

Linear-quadratic approximation

$$\max_{\Delta\phi} g^{\top}\Delta\phi \quad \text{subject to} \quad \frac{1}{2}\Delta\phi^{\top} F \Delta\phi \leq \delta, \qquad (9)$$

Issue: Even though we have selected an optimal step size for (9) in (10), we are not guaranteed that in (7), we

- actually get a descent in $L_{\text{TRPO}}(\phi)$,
- satisfy the constraint $\bar{D}_{\text{KL}}\left(\pi_{\text{old}} \| \pi_{\phi}\right) \leq \delta$.

Step length selection via backtracking. Define a *shrinkage parameter* $\alpha \in (0, 1)$ and

1. Get $\Delta\phi$ by solving (9).
2. Calculate candidate $\phi' = \phi_{\text{old}} + \Delta\phi$ for the next iterate.
3. Check the following two conditions (recall that $L_{\text{TRPO}}(\phi_{\text{old}}) = 0$):

$$(I) \quad L_{\text{TRPO}}(\phi') > 0\,? \qquad \text{and} \qquad (II) \quad \bar{D}_{\text{KL}}\left(\pi_{\text{old}} \| \pi_{\phi'}\right) \leq \delta\,?$$

4. **if** (I) and (II) are satisfied **then** set $\phi_{\text{new}} = \phi'$ and STOP. **else:** set $\Delta\phi \leftarrow \alpha\Delta\phi$ and go back to 2.

TRPO: Final algorithm

Input: Initial policy parameters $\phi^{(0)}$, trust-region bound δ, backtracking parameter $\alpha \in (0, 1)$, acceptance threshold $\epsilon > 0$.

For iterations $k = 0, 1, 2, \ldots$ until convergence:

1. **Data collection:** Run the current policy $\pi_{\phi^{(k)}}$ in the environment to collect a batch of trajectories $\mathcal{B} = \{(s_t, a_t, r_t)\}$.
2. **Advantage estimation:** Estimate the advantage values $A_{\phi^{(k)}}(s, a)$ for all sampled steps.
3. **Compute policy gradient (g):** Evaluate the standard policy gradient vector g using the collected samples:

$$g = \frac{1}{|\mathcal{B}|} \sum_{s,a \in \mathcal{B}} \nabla_\phi \log \pi_\phi(a|s)\Big|_{\phi=\phi^{(k)}} A_{\phi^{(k)}}(s, a)$$

4. **Compute Step Direction (x) via Truncated CG** (10 to 20 iterations) to solve the linear system $Fx = g$ for $x \approx F^{-1}g$.
5. **Maximal step size (β):** Compute $\beta = \sqrt{\frac{2\delta}{x^T F x}}$. Set the full proposed step direction: $\Delta\phi = \beta x$.
6. **Backtracking:** Find largest factor $\gamma = \alpha^j$ (for $j = 0, 1, 2, \ldots$) such that $\phi_{\text{new}} = \phi^{(k)} + \gamma\Delta\phi$ satisfies:

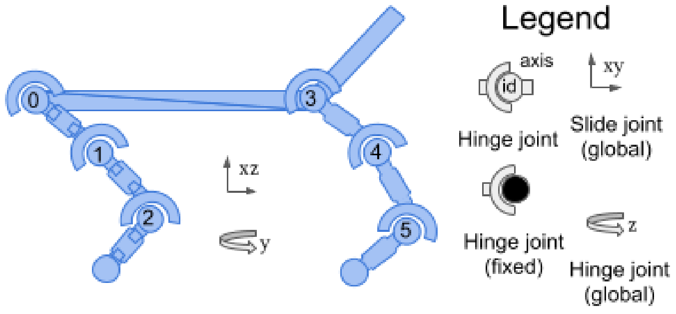
$$L_{\text{TRPO}}(\phi_{\text{new}}) > 0 \qquad \text{and} \qquad \bar{D}_{\text{KL}}\left(\phi^{(k)} \middle\| \phi_{\text{new}}\right) \leq \delta$$

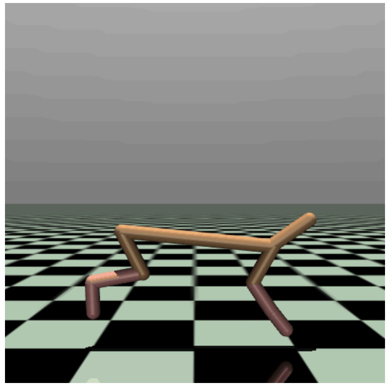
7. **Policy Update:** Update parameters to the safe, verified step: $\phi^{(k+1)} = \phi^{(k)} + \gamma\Delta\phi$ (or *reject* if no γ can be found).

Example: Half-Cheetah

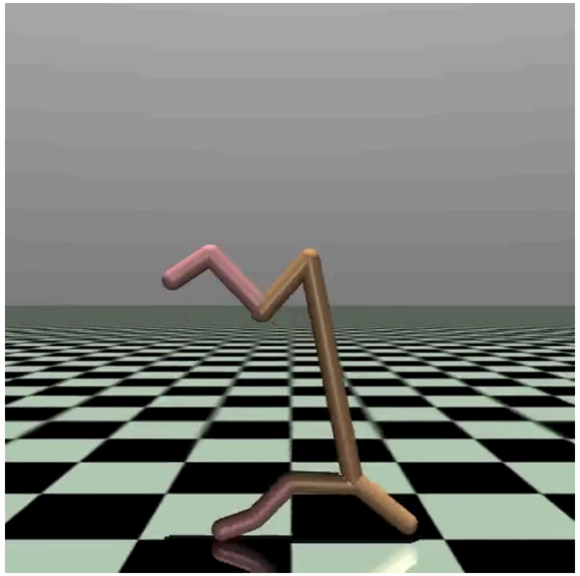
As an example, we study the Half Cheetah example from the MuJoCo library.

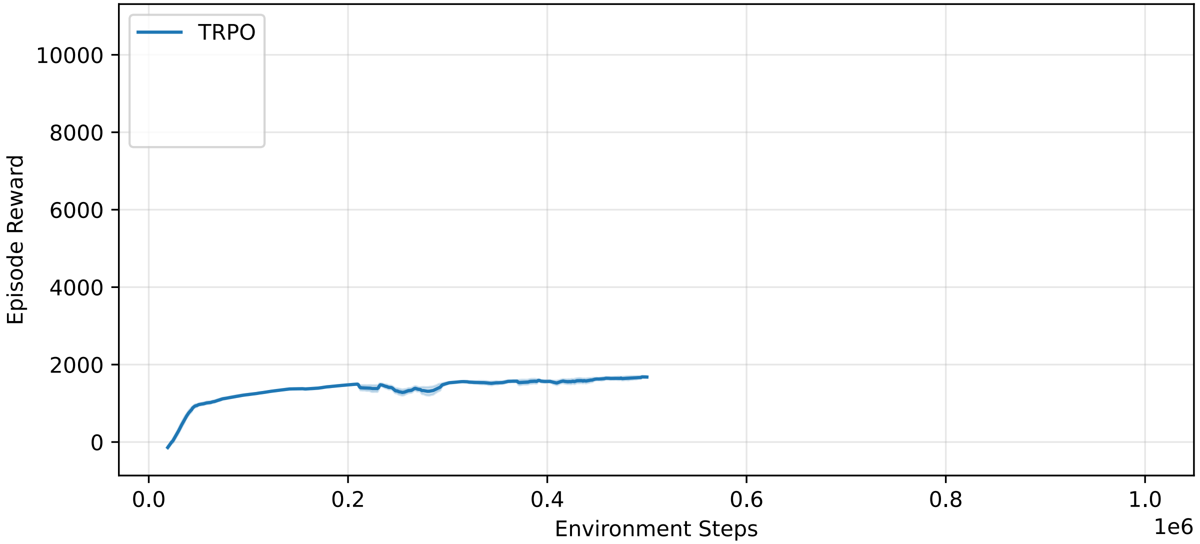
- $\mathcal{S}$: 17 states.
- $\mathcal{A}$: 6 actions.
- Reward: Forward-Cost - Control-Cost.





Results with TRPO




TRPO
10000
8000
6000
4000
2000
0
Episode Reward
0.0
0.2
0.4
0.6
0.8
1.0
1e6
Environment Steps

Proximal policy optimization (PPO)

Proximal policy optimization (PPO)

- TRPO solves $\max_\theta L(\theta)$ subject to $D_{KL}(\pi_{\theta_{\text{old}}}, \pi_\theta) \leq \delta$. To do this it requires:
  - second-order information (Fisher matrix)
  - conjugate gradient
  - a line search
  - KL constraint checks
- Most deep learning uses first-order optimizers (Adam, SGD). TRPO instead requires solving $F^{-1}g$, which means special optimization code rather than standard gradient descent.
- Because of the line search and constraint checks, each update is more expensive and harder to integrate into large-scale training pipelines. Researchers wanted something that:
  - keeps small policy updates
  - but uses simple gradient descent

Two possible approaches: ::: incremental 1. Turn constraint into penalty term (tuning constant is hard). 2. Clipped objective (the one people use)

Evolution of policy gradient methods

- policy gradient
- natural policy gradient
- TRPO: maximize surrogate objectivesubject to KL trust region
- PPO: replace hard trust region with clipped objective

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